JOURNAL OF COGNITION AND DEVELOPMENT, 7(1), 73–93
Copyright © 2006, Lawrence Erlbaum Associates, Inc.

You Must Remember This: Effects of Video and Photograph Reminders on 18-Month-Olds' Event Memory

Ellyn G. Sheffield and Judith A. Hudson
Rutgers University

The effect of video and photograph reinstatement on 18-month-olds' recall was assessed in 3 experiments. In Experiment 1, children learned to produce novel actions in a laboratory playroom, then returned 2 weeks later to either reenact the activities or watch a videotape of the actions. They were tested 10 weeks after reinstatement. Although children who had reenacted activities produced a substantial number of actions, children who watched a video presentation produced no more actions at test than children who had not watched the video. In Experiment 2, children watched a reminder video in the laboratory after 10 weeks and were tested either 15 min or 2 weeks later. Children in both groups showed evidence of reinstatement, but recall was significantly greater when testing was conducted 15 min after reinstatement. In Experiment 3, 10 weeks after training children either watched a reinstatement video or saw photographs of the event in their homes. They were tested 24 hr later in the laboratory. Effects of reinstatement were again found for video but not for photographs.

Recent investigations of memory in infants and toddlers have found that although children under the age of 3 years may not verbally report details from episodes, they remember information that is fundamentally similar to that which older children and adults verbally recall (Barr, Dowden, & Hayne, 1996; Bauer, 1996; Bauer & Hertsgaard, 1993; Bauer, Hertsgaard, & Dow, 1994; Bauer & Wewerka, 1997; Fivush & Hamond, 1989; Hudson & Sheffield, 1998). This growing body of research indicates that 1- and 2-year-old children retain well-ordered event memories over long periods of time, and their recall of event sequences is influenced by

Correspondence should be addressed to Ellyn G. Sheffield, Department of Psychology, Salisbury University, 1101 Camden Avenue, AW 406, Salisbury, Maryland 21801. E-mail: egsheffield@salisbury.edu

the same factors that influence recall in older children. For example, like preschool-age children, toddlers recall more when they are given specific cues, have repeated experience with an event, or when they are more active participants (Bauer & Wewerka, 1995).

To examine infants' and toddlers' memory capabilities, three types of paradigms have been used extensively: (a) response-contingency tasks, (b) deferred imitation tasks, and (c) elicited imitation tasks. In response-contingency tasks, infants learn to respond to a stimulus and produce a target behavior (e.g., making a mobile move by leg kicking, making a train travel around a track by depressing a lever). Depending on the age of the infant and the test paradigm, this training is accomplished during one or more sessions. If children as young as 8 weeks are trained twice, they remember and respond to a novel stimulus for up to 1 day (Greco, Rovee-Collier, Hayne, Griesler, & Earley, 1986). After this time period, infants do not respond to the stimulus as they had previously. With age, infants can remember these unique experiences for increasingly longer periods of time (Boller, Rovee-Collier, Borovsky, O'Connor, & Shyi, 1990). By 6 months of age, infants remember their training for up to 20 days; by 12 months they can remember the experience for up to 8 weeks; and by 18 months they can remember the experience for up to 13 weeks (Boller et. al, 1990; Rovee-Collier, 1995; Rovee-Collier & Hayne, 1987).

The ability for maturing children to remember for longer periods has also been studied using paradigms that are thought to be more closely associated with declarative memory. Using a deferred imitation paradigm, Meltzoff and Moore (1994) demonstrated that 24 hr after seeing a behavior, infants as young as 6 weeks of age could imitate human actions such as tongue protrusion. Older infants are capable of storing representations for significantly longer intervals. For example, although they had not had an opportunity to manipulate props, 4 months after being exposed to simple activities 14-month-old infants imitated an experimenter's actions when they were prompted with the original props (Meltzoff, 1995).

Using elicited imitation, Bauer and Mandler (1990) demonstrated that 16-month-olds remember action sequences for 6 weeks if they immediately imitate the actions during one session. Eighteen-month-olds can recall activities they learned in a single trip to a laboratory playroom for over 2 months, and 24-month-olds remember activities for well over 4 months (Hudson & Sheffield, 1998). Bauer, Hertsgaard, and Dow (1994) demonstrated that 2-year-olds recalled information learned during a laboratory experience occurring 8 months in the past. In this experiment, children learned to perform a series of multistep activities, were tested 1 week later and again 8 months later. The experienced children's performance was compared to a group of age-matched naive children who came to the laboratory once to play with the props. In long-term testing, the 2-year-olds who had originally experienced the laboratory events performed action sequences significantly more often than children in the control condition.

There is substantial evidence that like older children, infants' and toddlers' memories for unique events can be extended by providing them with reminders, either prior to or during a test situation (Bauer, Wenner, Dropik, & Wewerka, 2000; Hudson & Sheffield, 1998; Rovee-Collier, 1995; Rovee-Collier & Hayne, 1987). Bauer et al. found that very young children's memories for event sequences to which they had been previously exposed improved when verbal labels for activities were provided by the experimenter during the final test. The recall measure used in this study consisted of children's performance of action sequences. This finding suggests that young children's memories are being assisted by verbal cues if they are presented at the time of testing.

Reminders may also be presented prior to test situations, at varying intervals after initial learning has taken place. This process is known as reinstatement. In reinstatement paradigms, reminders include exposure to part of a previously experienced event where the information contained in the reminder is not complete enough to cause new learning (Campbell & Jaynes, 1966). Reinstatement effects vary, depending on the age of the child, the amount and type of information the reminder provides, the time at which the reminder is provided, and the medium in which the reminder is presented (Hartshorn, 2003; Hildreth, Sweeney, & Rovee-Collier, 2003; Howe, Courage, & Bryant-Brown, 1992; Rovee-Collier, 1995).

In reinstatement experiments with toddlers and preschool children, children have been reminded of events by (a) seeing a portion of an event, (b) reenacting a subset of activities contained in an event, (c) watching an experimenter model activities that they once experienced, or (d) viewing objects with which they were initially trained (Galluccio & Rovee-Collier, 1999; Howe et. al., 1992; Hudson & Sheffield, 1998; Sheffield & Hudson, 1994). In all of these cases, regardless of age, when children are provided with a reminder of the event before testing, they remember more than children who were not provided with reminders.

Although children under 3 years can retain event memories for several months, very young children still forget a considerable amount even minutes after an experience and their memories are more limited in duration than are older children's (Bauer et al., 2000; Brainerd, Reyna, Howe, & Kingma, 1990; Hartshorn et al.,1998; Howe, Kelland, Bryant-Brown, & Clark, 1992, Rovee-Collier, 1999). Older children's memories may endure for longer periods because they can use a wider range of representational reminders, such as language, to extend their recall (Hudson, 1990; Nelson, 1993). By initiating and engaging in conversations about past events, older children's memories for past episodic events may be activated and reinstated. In contrast, results from two studies indicate that conversation has little reinstatement effect on very young children's recall. In these studies, 2-year-olds who engaged in conversations about a past event did not recall more than children who were not verbally reminded (Hudson, 1990, 1993).

Other types of representational reminders, such as photographs, may be more effective in reinstating young children's memories (Deocampo & Hudson, 2003). In Deocampo and Hudson's study, 2-year-olds watched an experimenter demonstrate multistep activities and were later reinstated by seeing photographs that depicted partially completed activities. Both 24- and 30-month old children who saw the photographs remembered more activities than children who did not. Therefore, for 2-year-olds, photographic reminders provided reinstatement. However, results from this study also indicated that 24-month-olds whose memories were reinstated with photographs remembered only 30% of the target actions. In contrast, 30-month-olds remembered 70% of the targets. Thus, although there was evidence of reinstatement for the younger children, older children demonstrated better recall after reinstatement. This indicated that whereas photographs successfully reinstated young 2-year-olds' memories, this kind of reinstatement benefited older 2-year-olds' memories to a far greater extent.

Research suggests that children even younger than age 2 years can learn from and remember information contained in a representational medium but that successful performance may be affected by both the complexity of the representation and the task that the children are being asked to perform. Using deferred imitation, Meltzoff (1988) demonstrated that 24 hr after exposure, 14-month-olds could recall one-step actions modeled on television by an experimenter. However, Barr and Hayne (1999) found that 24 hr later infants under 15 months had difficulty imitating multistep tasks originally presented over video. In this study, 12-, 15-, and 18-month-olds were shown activities either over videotape or in a live demonstration and were tested 24 hr later on their ability to reproduce the activities. At all ages, infants who were shown the activities in a live demonstration produced more target actions than children who had either seen a video demonstration or who had not participated in the demonstration. Twelve- and 15-month-olds did not show evidence of recall when their initial experience was presented via videotape. Eighteen-month-olds who watched the video demonstration were more successful, remembering significantly more target actions than age-matched controls.

The preceding research raises several interesting questions concerning the effects of representational reinstatements on young children's memories. Can 18-month-old children use reminders contained in representational media to reinstate their memories? If so, in comparison to nonrepresentational reinstatements (i.e., physical reenactments), how effective might these reminders be? In this study we investigated whether very young children's memories can be reinstated by exposure to information contained in representational media and whether this reinstatement is as enduring as other kinds of physical reinstatements. In Experiments 1 and 2 we examined the effects of video reminders on children's event memory, and in Experiment 3 we examined the effects of both video and photograph reminders.

EXPERIMENT 1

In this experiment, we examined the effect of a video reinstatement on 18-month-olds' memories for actions learned during a novel event, a trip to a laboratory playroom. Eighteen-month-olds were studied for several reasons. Prior research (Hudson & Sheffield, 1998; Sheffield & Hudson, 1994) showed that reenactment and live modeling were effective in reinstating 18-month-olds' event memories. This research provided comparison data to test for the effects of other kinds of reminders and also established effective time intervals for reinstatement testing at this age. In addition, research on children's use of photographs showed that 24-month-olds could use photographs to reinstate their memories (Deocampo & Hudson, 2003), and this point established an upper bound on when children could be expected to use representational reminders. Thus, 18 months served as a logical starting point to test the effects of representational reminders on long-term recall.

Using an elicited imitation paradigm, children were shown eight novel activities and were encouraged to imitate them. Children in a video reinstatement condition returned to the laboratory 2 weeks later and watched a video simulation of the activities, and then returned for testing 10 weeks after the reminder session (or 12 weeks after initial training). This timing was chosen based on prior research that showed that in a similar elicited limitation paradigm, 18-month-olds could not recall most of their training after 10 weeks without reinstatement, but with reinstatement they could recall a majority of activities from their initial experience (Sheffield & Hudson, 1994). Because results from studies suggest that returning to the original training context can reactivate infants' memory for a previously learned response (Butler & Rovee-Collier, 1989; Rovee-Collier, Griesler, & Earley, 1985), we included a second video reminder condition to test whether returning to the laboratory context would reinstate children's memory of their initial training experience. In this condition, children were trained to imitate activities and returned to the laboratory 2 weeks later to watch an unrelated video segment. They returned 10 weeks later for testing.

Method

Participants

A total of 50 children, age 18 months, participated. Children were recruited from birth announcements and word of mouth. None of the children had any prior contact with the laboratory, experimenters, or activities. Two children failed to produce the actions during training, and 2 children were prevented from testing due to scheduling problems. Children of the same age and gender replaced these

TABLE 1
Schedule of Initial Experience, Reminder, and Recall Sessions

Condition	*Initial Learning*	*Reinstatement Type Timing (Weeks)*	*Recall (Weeks)*
Experiment 1			
Video reinstatement	Day 1	Video 2	12
Video reinstatement control	Day 1	Video 2	12
Live re-enactment	Day 1	Re-enact 2	12
No reminder	Day 1	—	12
Video training 2–12	—	Video 2	12
Experiment 2			
Video reinstatement 10–10	Day 1	Video 10	10
Video Reinstatement 10–12	Day 1	Video 10	12
Video training 10–10	—	Video 10	10
Experiment 3			
Photograph reinstatement	Day 1	Video 10	10
Video reinstatement	Day 1	Photo 10	10
Photograph training	—	Video 10	10
Video training	—	Photo 10	10
No reminder	Day 1	—	10

children. Children were predominately from Caucasian, middle-class families. A questionnaire completed by mothers indicated children watched an average of 1.75 hr of television per day (range = 0–4 hr). This indicated that, like 18-month-olds in other studies (Lemish & Rice, 1986), the majority of our participants had been regularly exposed to television.

Participants were randomly assigned to one of five conditions as shown in Table 1 (five girls and five boys in each condition). Table 2 lists participants' mean ages and range of ages for each condition.

Design

The performance of children who viewed either a video simulation of the event (the video reinstatement condition) or a video containing no pertinent information (the video reinstatement control condition) was compared to several additional conditions. The first condition (the live reenactment condition) consisted of children who were given an additional opportunity to physically enact the activities in a second training session. Past research indicated that 18-month-olds who reenacted activities 2 weeks after initial training recalled the majority of their original experience after 10 weeks, whereas children who did not reenact the activities failed to reproduce most of the activities in delayed recall testing (Hudson & Sheffield, 1998). However, because many of the activities presented in this experiment were new, testing for replication of this finding seemed prudent.

TABLE 2
Mean Ages and Age Rangers of Participants in Experiments

Condition	*Mean Age (Months)*	*Age Range (Months)*
Experiment 1		
Video reinstatement	18.0	17.25–18.75
Video reinstatement control	18.0	17.25–18.75
Live re-enactment	18.0	17.25–18.75
No reminder	18.5	17.5–19.0
Video training 2–12	18.25	17.5–19.5
Experiment 2		
Video reinstatement 10–10	18.0	17.25–18.75
Video reinstatement 10–12	18.25	17.5–19.25
Video training 10–10	18.5	
Experiment 3		
Photographic reinstatement	18.0	17.75–18.75
Video reinstatement	18.25	17.5–19.25
Photograph training	20.5	19.5–21.5
Video training	20.4	19.75–21.25
No reminder	18.5	17.0–19.0

The second control condition (the no-reminder condition) provided a measure of recall with no reinstatement. Children in this condition were trained to imitate the activities and were simply tested 12 weeks later. A final group, the video training 2–12 condition, provided a measure of children's ability to imitate the actions shown on the video when they had not been shown the activities in a live training session. These children did not participate in the training session but simply watched the video reminder and were tested for recall 10 weeks later. To conclude that reinstatement was effective, children in the video reinstatement, video reinstatement control, or live reenactment groups needed to demonstrate significantly greater recall than children who were shown the video as a training device or received no reminder.

Apparatus

Activities. As described in Table 3, eight novel, developmentally appropriate activities were selected. Several of these activities were adapted from previous studies (Hudson & Sheffield, 1998; Sheffield & Hudson, 1994), and several were newly created. Each activity consisted of 2 specific target actions for a total of 16 target actions. Activities were designed to be enjoyable for children and included actions that could not be easily inferred from simply observing or manipulating the props. Several actions were performed with objects that were hidden from view so children rarely produced the target actions without viewing a demonstration. The performance of children in the video training 2–12 group who

TABLE 3
Descriptions of Activities and Target Actions

Activity	*Description*
Feed the fish	*Find fish food* in drawer and *shake it* over fish tank.
Make a rattle	Rotate cylinder on holder until opening shows. Take rings off holder, *put in cylinder*, and *shake "rattle."*
Make a crown	*Attach two plastic jewels* to crown with Velcro and *put crown on head*.
Mickey Mouse	*Find "special friend" (Mickey)* from an array of three stuffed toys. Take Mickey off trunk, and *pull ring* (on back) to make toy talk.
Clown	*Take stuffed clown out of toy chest*, put in the rocking chair, and *rock him*.
Pop-up toy	*Pull hammer out* of side of toy. *Hit shapes* on the bottom to make plastic pieces "pop" out.
Squeaky toy	Take red box off shelf (select from three colored boxes), *open box*, take out toy, and *squeeze toy* to make noise.
Seesaw	Take ball out of holder and *place ball on one end* of seesaw. *Push the other side*, making the ball fly out.

Note. Target actions of italicized.

did not participate in training provided a measure of performance without prior live demonstration.

Videos. Three 3.5-min videos were produced. Two were created for the video reinstatement condition: One included a female child (shown to female participants) and one included a male child (shown to male participants). These videos included complete information from all eight activities presented in the same order in which children were trained. Children saw an experimenter introduce herself to a child and help the child recreate all actions for each activity, one by one. The experimenter on the video provided a narrative while the activity was being demonstrated (e.g., "Look, here's my fish. Do you remember what we did with the fish? That's right, there is the fish food, in the drawer. Now take the food and shake it on top of the fish tank"). This script was the same as the narrative used by experimenters during reenactment sessions. The third video, a video reinstatement control, consisted of a 3.5-min segment from *Sesame Street,* including a song and dance routine by the Muppets about pigeons.

Procedure

For initial training and final testing, the props associated with the target activities were placed at specific locations in the playroom, accessible to children. Other toys that were not used in the target activities were also placed around the room. All training sessions and final test sessions lasted between 20 and 30 min, and all reinstatement sessions lasted between 7 and 10 min. Parents were instructed to remain with their children during all visits but were asked to refrain from providing verbal instruction or other assistance to their children.

Training session. The experimenter demonstrated eight target activities, one at a time, using props as described in Table 3. She modeled the target actions for each activity and encouraged the child to immediately imitate them. All children produced the target actions at least once before the next activity was introduced. Activities were presented in the same order as shown in Table 3, and the experimenter remained the same throughout all sessions.

Reinstatement session. All reinstatement sessions took place 2 weeks after the training session. Depending on condition assignment, during the reinstatement session, children (a) viewed a video simulation of a training session, (b) watched a taped segment from Sesame Street, or (c) reenacted the original activities. Children in the video training 2–12 condition were matched in age to children in the other conditions (they were 18 months and 2 weeks at the time of their first session).

In both the video reinstatement and video reinstatement control conditions, props associated with the eight activities were removed from the playroom. Distracter props were left in the playroom for children to examine and manipulate. A 48-cm color television and VCR on a table were positioned in the playroom diagonally across from a couch. Children were permitted approximately 5 min of free time in the playroom prior to watching the video simulation. Children were then encouraged to sit with their parent and watch the videotape. The experimenter directed children's attention to the television by saying "Look at the TV" or "Look at the little girl/boy." Specific names of the activities or target actions were not mentioned. If children became distracted during the video presentation, the tape was paused until they resumed watching it. Sessions lasted 7 to 10 min.

In the live reenactment condition, all of the original props were present in the playroom. The experimenter directed children's attention to each activity, and children were encouraged to produce the target actions as they had in the first training session. If children did not produce the target actions, the experimenter modeled the target actions again. All children performed the target actions for each activity once before the next activity was introduced. To match children's total time spent in the laboratory playroom in the live reenactment condition to time spent in the video reinstatement conditions, all reinstatement sessions lasted approximately 10 min. These sessions were shorter than the original session because children required less time to warm up to the Experimenter and understand experimental procedures.

Test session. As shown in Table 1, children in the video reinstatement, video reinstatement control, live reenactment, and no-reminder conditions returned to the laboratory 12 weeks after their first visit. Children in the video training control conditions returned 10 weeks after their first visit. The playroom was arranged as it had been during initial training. Children were first allowed 5 min of free exploration in the room when they were not directed toward any particular ac-

tivity. Then the experimenter directed children's attention to each of the activities and encouraged children to perform the target activities, without providing specific instruction. For example, the experimenter asked, "Can you feed the fish?" but did not show children where the food was located and asked "What can you do with this?" while pointing to the seesaw apparatus.

Coding

All test sessions were videotaped for the purpose of subsequent coding. During the test session, spontaneous recall was scored as any actions children produced during the 5-min play period before the experimenter started prompting them. Cued recall was scored as any actions children produced in response to the experimenter's prompting. There were 2 target actions for each activity for a total of 16 target actions. Children were credited with 1 point for each target action that they produced. Thus, for each activity, children received a score from 0 to 2. Coding was done either in real time behind a one-way mirror or from videotape. The first observer, unaware of condition, coded all of the actions as children were performing them at final test. A second experimenter, unaware of condition, coded tapes of 2 participants in each condition. Intercoder agreement was computed by determining the total number of possible scores for each child (16 at the test session). This total was divided by the number of scores on which both coders agreed. Intercoder agreement was 99% for individually produced target actions.

Results

A 2 (Gender) × 5 (Condition: video reinstatement, video reinstatement control, live reenactment, no reminder, video training 2–12) analysis of variance (ANOVA) was performed both on the total number of spontaneous actions produced during the final test session and the total number of cued actions. This analysis produced no significant main or interaction effects for gender, so this variable was not included in subsequent analyses.

Activity Analysis

An analysis was performed to determine whether some target actions were performed more than others due to the intrinsic appeal of specific activities. A 5 (Condition) × 8 (Activity) ANOVA was performed with condition as a between-participants factor and activity as a within-participants factor. This analysis yielded a main effect of condition, $F(4, 399) = 18.40$, $p < .001$, and a main effect of activity, $F(7, 399) = 5.37$, $p < .001$, suggesting that some activities were easier to encode or remember or both, or that some were more appealing than others. However, there was no significant interaction, suggesting that performance of specific activities was not a function of condition assignment.

TABLE 4
Mean Spontaneous and Cued Scores, Experiment 1

Condition	*Spontaneous Productions*	*SD*	*Cued Productions*	*SD*
Video reinstatment	2.8	1.47	2.7	1.25
Video reinstatement control	2.1	1.10	2.0	1.49
Live re-enactment	5.9[a]	2.68	3.9	2.33
No reminder	2.6	1.42	2.0	1.63
Video training 2–12	.9	1.10	2.2	1.47

[a]Different from all other conditions.

Effects of Reinstatement: Spontaneous and Cued Responses

The mean number of target actions produced spontaneously and in response to cue by children in each condition are shown in Table 4. A 5 (Condition) × 2 (Cue Type) multivariate ANOVA was performed with condition as a between-participant factor and cue type as a within-participant factor. The Wilks' lamdba test yielded a significant term $F(8, 88) = 6.73$, $p < .001$. Univariate ANOVAs for both spontaneous and cued responses were then conducted to determine which of the variables and factors caused significance. With regard to spontaneous production, the ANOVA yielded a significant effect, $F(4, 49) = 12.39$, $p = .001$. Fisher's LSD Multiple-Comparison post hoc tests ($p < .05$) indicated that children in the live reenactment condition performed significantly more target actions than all other groups in spontaneous recall, whereas the video reinstatement condition performed no more targets than children in the control groups. With regard to cued production, the ANOVA did not yield a significant effect, $F(4, 49) = 2.29$, $p = .0748$, indicating that children in all conditions performed similarly. Taken together, these analyses indicate that whereas children in the live reenactment condition performed significantly more target actions than children in all other conditions, children in the video reinstatement condition performed no more actions than children in the control conditions.

Discussion

In Experiment 1 we did not find evidence that a video presentation reinstated 18-month-olds' memories when the reminder was administered 2 weeks after training and children were tested 10 weeks after the reminder session. Only children in the live reenactment condition produced more target actions than children in control conditions. Furthermore, children who reenacted the event produced over half of the target actions ($M = 9.8$), indicating that their memory for the event

was fairly robust. Although this finding replicated past studies showing that reenactment successfully reminded children of their past training (Hudson & Sheffield, 1998), it is important to note that we do not understand fully why these multiple training trials were so effective in reinstating children's memory, and why the video reinstatement was not. It is possible that actively participating in our laboratory experience twice afforded children the maximum opportunity to encode the various components of the event (i.e., activities, context, etc.). Or, it may be that children simply received more exposure to the props during reinstatement by virtue of having the activities modeled more effectively (i.e., more times, more slowly, with more exaggeration) than they did during the video session. Finally, children may have been engaged and encouraged by the experimenter in ways that are unique to a live modeling session.

It was also curious that children in the video reinstatement condition failed to produce more target actions than children in the control conditions, either spontaneously or in response to cueing, since deferred imitation research showed that children at 18 months successfully remembered information they had learned from a video presentation (Barr & Hayne, 1999). There are several plausible explanations for the video's lack of effect. First, the video reminder may have had no effect because children either did not attend to the video, or the video actions went by too quickly to be well understood. Although we believe that lack of attention could have affected the reinstatement process, we were careful to pause the video and redirect children's attention to the display if we saw their attention waning.

Second, the video reminder may have been ineffective because children could not transfer information from their original experience with 3-dimensional objects to a 2-dimensional reinstatement and transfer it back to a 3-dimensional test condition. This explanation assumes that children were having difficulty using representational information contained in the 2-dimensional video to reinstate their memory for real-world events.

Finally, it is possible that the information contained in the video did act as a reinstatement, but that the effects of the reminder did not endure for 10 weeks. Although the effects of live reenactment are long-lasting, it is possible that information contained in other kinds of representational media, such as video, may be effective in the short term, but less effective over long periods. This explanation is supported by Barr and Hayne's (1999) deferred imitation research demonstrating that 15-month-olds were able to remember an activity for 24 hours when it was modeled by an experimenter, but evidenced forgetting over the same delay when the activity was presented on video.

EXPERIMENT 2

In order to test the possibility that reinstatement effects did not endure over a 10-week delay period, we conducted a second experiment in which the video rein-

statement occurred at the end of the retention interval, 10 weeks after training, when children would have forgotten the event without being reminded of it. This timing was chosen in order to determine if reinstatement would be effective after a very long period, when children's memory for the event was functionally forgotten. Past research with 18-month-olds in a similar paradigm showed that children produced no more target actions than age-matched controls after a 10-week interval if they received no reinstatement (Sheffield & Hudson, 1994). Children were tested 15 min after the reinstatement (the video reinstatement 10–10 condition). Additionally, to chart the course of reinstatement effectiveness over time, another group of children were tested 2 weeks after they watched the video (the video reinstatement 10–12 condition). Finally, a group of untrained children were tested 15 min after watching the training video (the video training 10–10 condition) to ensure that production of target actions was a reflection of reinstatement, and not merely a successful demonstration of deferred imitation.

Method

Participants

Thirty 18-month-olds (14 females and 16 males) participated in this study. Participants were recruited as in Experiment 1. Children were predominately from Caucasian, middle-class families. A questionnaire completed by mothers indicated children watched approximately the same amount of television as children in Experiment 1.

As described in Table 1, 10 participants were randomly assigned to each of three conditions, the video reinstatement 10–10 condition, the video reinstatement 10–12 condition, and the video training 10–10 condition. Table 2 shows their average ages and age ranges.

Design and Procedure

Training session. All training was performed as in Experiment 1 and lasted between 20 and 30 min. Children in the video training 10–10 condition did not receive initial training.

Reinstatement session. Reinstatement sessions took place 10 weeks after the training session for children in the reminder conditions and consisted of a session in which children watched the same video simulations used in Experiment 1. Reinstatement sessions lasted from 7 to 10 min. Children in the video training 10–10 condition were matched in age to children in the reinstatement conditions. Thus, they were 18.5 months at the time of reinstatement.

Test session. After viewing the video, children in the video reinstatement 10–10 and the video training 10–10 conditions left the laboratory, were given a

snack, and took a walk in another part of the building for several minutes. Then they returned to the playroom for testing approximately 15 minutes after viewing the videotape. Children in the video reinstatement 10–12 condition returned to the lab for testing two weeks after watching the video. As in Experiment 1, the playroom was arranged as it had been during initial training. After 5 min of free play in the room, the experimenter directed children's attention to each activity and provided them with orienting prompts.

Coding

As in Experiment 1, test sessions were videotaped for the purpose of subsequent coding. One observer coded all of the sessions in real-time and a second experimenter, unaware of condition, coded tapes of two participants in each condition. As in Experiment 1, inter-coder agreement was 99% for individually produced target actions.

Results and Discussion.

Data from children in the video reinstatement group from Experiment 1 were included in the analyses as the video reinstatement 2–12 condition to test for the duration effects of the video reinstatement. A no reminder control group was not included because comparisons to the video reinstatement group who received initial training, viewed a video at 2 weeks, and returned for testing 10 weeks later provided an even more stringent control comparison. A preliminary 2 (Gender) × 4 (Condition: video reinstatement 2–12, video reinstatement 10–10, video reinstatement 10–12, video training 10–10) ANOVA was performed on the number of target actions produced during the test session. This analysis showed no significant main or interaction effect for gender so this variable was excluded in subsequent analyses. Because the same activities were used in all experiments, additional activity analyses were not conducted for Experiments 2 and 3.

The mean numbers of target actions produced in each condition are shown in Table 5. A 4 (Condition) × 2 (Cue type) multivariate ANOVA was performed with

TABLE 5
Mean Spontaneous and Cued Scores, Experiment 2

Condition	*Spontaneous Productions*	*SD*	*Cued Productions*	*SD*
Video reinstatement 10–10	5.5[a]	2.27	4.6[a]	1.89
Video reinstatement 10–12	2.6	2.06	4.7[a]	1.63
Video training 10–10	1.4	1.07	2.7	1.94
Video reinstatement 2–12[b]	2.8	1.47	2.7	1.25

[a]Different from Video Reinstatement 2–12 and Video Training Control 10–10.
[b]Data for this condition are from Experiment 1.

condition as a between-participant factor and cue type as a within-participant factor. The Wilks' lamdba test yielded a significant term, $F(6, 70) = 5.83$, $p < .001$. Univariate ANOVAs for both spontaneous and cued responses were then conducted to determine which of the variables and factors caused significance. With regard to spontaneous production, the ANOVA yielded a significant effect, $F(3, 39) = 9.39$, $p = .001$. Fisher's LSD multiple comparison post hoc tests indicated that children in the video reinstatement 10–10 condition performed significantly better than all other groups in spontaneous recall, whereas the video reinstatement 10–12 condition performed no better than children in the video reinstatement 2–12 group. With regard to cued production, the ANOVA also yielded a significant effect, $F(3, 39) = 4.93$, $p = .001$. Fisher's LSD post hoc tests ($p < .05$) indicated that children in the video reinstatement 10–10 and video reinstatement 10–12 conditions performed significantly better than children in the video reinstatement 2–12 condition and video training control 10–10, suggesting that reinstatement had an effect after 2 weeks when children were cued to particular activities. These findings indicate that viewing a video reminder immediately prior to recall testing effectively reminded children of the past training event. Although reminder effects were still found for children in the video reinstatement 10–12 condition as evidenced by their significantly higher level of cued production, their spontaneous recall was no better than children in control conditions. This finding suggests that the effects of the video reminder diminished over the 2-week interval.

In contrast to these results, Experiment 1 showed that reenactment was effective even 10 weeks after reenactment. Thus, although video was an effective reminder for 18-month-olds, the effects were not as long lasting as physically reenacting the events.

EXPERIMENT 3

Results from Experiment 1 showed that watching an unrelated video in the laboratory context did not enhance children's recall. Nevertheless, the unrelated video was shown 2 weeks after training at a time when the simulation video was also ineffective. Further, in both Experiments 1 and 2, it is possible that returning to the original training context served as the reminder and that reinstatement results were not due to watching the video but were merely an effect of returning to the laboratory context at the end of the retention interval. Research with infants has found that reinstating the original training context is critical for reinstatement to occur (Butler & Rovee-Collier, 1989; Rovee-Collier, et al., 1985) but other research with older infants suggests that context may be less of a factor (Klein & Melzoff, 1999). Although the original props were not available during video viewing, children nevertheless experienced coming to the laboratory building and entering the laboratory area prior to video viewing, which could have been encoded as part of the

larger training context. Therefore, to separate the contribution of video and context in the reinstatement process, in Experiment 3 the video reminder was shown to children in their homes 1 day before recall testing. Thus, all children visited the laboratory only twice: once for original training and once for final testing.

We were also interested in determining whether photographs accompanied by running narration could reinstatement 18-month-olds' memory. Deocampo and Hudson (2003) found that showing 24-month-olds photographs of a training event reminded them of their past experience. Additionally, a study by Sheffield (2004) found that memories of 18-month-olds who watched a video containing only object information about activities were successfully reinstated, in spite of the fact that no action information had been provided. Given these findings, we hypothesized that photographs might effectively reinstate 18-month-olds in this paradigm, similar to watching a video.

Method

Participants

Fifty 18-month-olds (26 girls and 24 boys) participated in this study. Children were recruited as in Experiments 1 and 2 and were predominately from Caucasian, middle-class families. As shown in Table 1, participants were randomly assigned to one of five conditions (10 children in each condition): photograph reinstatement, video reinstatement, photograph training, video training, and no reminder. Children in the reminder only conditions were matched in age to children in the reinstatement conditions at the time of reinstatement. Table 2 lists mean ages and age ranges.

Design and Procedure

Training session. All training sessions were performed as in Experiment 1 and lasted between 20 and 30 min. Children in the video training and photograph training did not receive initial training.

Reinstatement sessions. Reinstatement sessions took place 10 weeks after the training session. All reinstatement sessions took place in children's homes. For children in the photograph reinstatement and photograph training conditions, children were shown two 12.5 cm. × 17.5 cm. photographs for each of the eight activities demonstrated in the training session. Photographs for each activity were displayed on a single page in a photograph album. As children looked at the photographs, an experimenter described the action. For example, while viewing a photograph of a child feeding the fish, the experimenter said, "Look, the girl is feeding the fish. Do you remember when you did that? First you took the fish food out of the drawer, and then you shook it over the tank to feed the fish." In the photo-

graph training condition, the experimenter said, "Look the girl is feeding the fish. You are going to do that tomorrow when you come to my playroom. First you are going to take fish food out of the drawer, and then you are going to shake it over the tank and feed the fish." Children in the video reminder conditions watched the same video simulation of the activities as used in Experiments 1 and 2, but they watched them in their homes. Reinstatement sessions lasted from 7 to 10 min. Narration in the photograph group was matched in content and duration to the video script. Thus, the most salient difference between the video and photograph groups was the media that presented the visual representation.

Test sessions. Final testing was conducted 24 hr after reinstatement. This timing was chosen for logistical reasons and to ensure that children were not producing more activities merely because they were behaviorally aroused by the video reminder, as originally suggested by Spear and Parsons (1973). As in Experiments 1 and 2, the playroom was arranged as it had been during initial training. Testing sessions lasted from 20 to 30 min. Children were permitted 5 min of free play with the props before the experimenter directed children's attention to the props associated with each activity using general, orienting prompts.

Coding

As in Experiments 1 and 2, test sessions were videotaped for subsequent coding. One experimenter coded all of the tapes, and a second coder who was unaware of participants' condition independently coded tapes from 2 participants in each condition. Intercoder agreement was 97% for individually produced target actions.

Results and Discussion

A 5 (Condition) × 2 (Cue Type) ANOVA was performed with condition as a between-participant factor and cue type as a within-participant factor. This analysis yielded a main effect of condition, $F(4, 99) = 12.\ 57$, $p < .001$, but no effect of cue type and no interaction. Because there was no effect of cue type, spontaneous and cued actions were combined into a total score and are shown in Table 6. Fisher's LSD multiple comparison post hoc tests indicated that children in the video reinstatement condition performed significantly more actions than children in all other conditions. Thus, the video was effective in reinstating children's recall when shown to children in their homes but the photograph with accompanying narration was not effective as a reminder. Children in the photograph reinstatement condition recalled no more actions than children in the control conditions, indicating that reinstatement did not occur.

TABLE 6
Total Productions, Experiment 3

Condition	*Productions*	*SD*
Photograph reinstatement	6.1	2.47
Video reinstatement	10.9[a]	1.91
Photograph training	3.5	1.58
Video training	5.0	1.83
No reminder	4.5	1.90

[a]Different from all other conditions.

GENERAL DISCUSSION

The results from these experiments indicate by 18 months of age, children's event memory can be reinstated with at least one representational reminder, a videotaped presentation. Children do not need to physically repeat an activity or view actual objects associated with the activity to be reminded of it. Merely watching a videotape of a past event can also be an effective reminder. Further, children can be reminded of the past whether they see the video information in the environment in which they originally experienced the event or in another environment such as their homes. This development represents a departure from the ways in which younger children's recall is reinstated, that is, through reenactment or by viewing live demonstrations of actions using the same props as in the original event (Rovee-Collier & Hayne, 1987; Sheffield & Hudson, 1994). Children's abilities to use reminders across reminding contexts provide new opportunities for reinstatement to occur and may increase the likelihood that event memories are reinstated.

However, video reinstatement effects do not appear to be as long lasting as effects of physical reenactment. When shown a video of an event experienced 10 weeks before, children who were tested 15 min later showed higher levels of recall than children who were either not shown the video or who viewed the video without benefit of training. When tested 2 weeks later, their recall had already decreased and by 10 weeks they did not demonstrate recall at all. In contrast, research by Hudson and Sheffield (1998) showed that reenacting activities 8 weeks after training significantly improved 18-month-olds' recall when tested even up to 6 months after the reenactment session. Thus, watching a video of an event may be effective in reminding 18-month-olds of their past experience, but it is clearly not equivalent to actually reexperiencing the event.

The finding that children can use video information to remind them of a past experience at 18 months adds to a growing body of research that shows that very young children can learn and remember activities presented on video (Barr & Hayne, 1999; Meltzoff, 1988). Results also suggest that the ability to use video as a reminder precedes the ability to use photographs as reminders. In contrast to

watching a video demonstration, children who saw a photograph with the same accompanying narration failed to recall their initial training experience. Interestingly, this result contrasts with research demonstrating that children can be successfully reinstated by a video demonstration, even when it includes no action information (Sheffield, 2004).

This discrepancy raises questions that require further exploration. It is possible that the photograph reminders used in Experiment 3 were not effective in reinstating children's recall because they did not provide enough event information (Deocampo & Hudson, 2003). Whether a reminder is effective in reinstating a memory may be determined by the number of cues in the reminder that match a child's memory representation, rather than the medium in which the reminder information is presented. Reexperiencing an event in its entirely, watching someone else recreate the event, or watching someone else recreate the event on video all include exposure to all of the original event elements. In contrast, photograph cues contain less information about the event and may therefore provide fewer memory cues. Further, it is possible that photographs and narration may have some reinstating effects but that they last less than 24 hr. If this were true, in this paradigm we would not see evidence of reinstatement. Additional research testing immediate retention after seeing photographs would be useful to determine whether photographic reinstatement is effective but fleeting.

The results of this investigation show that for very young children the way in which a reminder is presented has a great deal to do with whether their memories will sustain over long periods. Live reenactment was clearly the only type of reminder that endured. Video provided reinstatement, but within weeks children's production of target actions was dramatically reduced. If one of the important components of recalling the past over very long periods is whether a reinstatement will provide an enduring memory, it is likely that very young children's long-term memory will not be assisted by representational reminders, such as videos, photographs, or verbal narration. Thus, one of the important components of remembering our earliest events, meaningful and varied reinstatement opportunities, may be unavailable to very young children, reducing the likelihood that early memories will be available into our adult years.

ACKNOWLEDGMENTS

The authors thank the children and parents who participated in this research. We thank Mimi Ryder for her contributions to Experiment 1 and Carolyn Zimmer for her contributions to Experiment 3. The assistance of Michelle Acala, Joseph Lieberman, Julia Loykster, and Dena Young in data collection and coding are also appreciated. Portions of these findings were presented at the meeting of the Society for Research in Child Development, April 1995.

REFERENCES

Bauer, P. J. (1996). What do infants recall of their lives? Memory for specific events by one-to two-year-olds. *American Psychologist, 51,* 29–41.

Bauer, P. J., & Hertsgaard, L. A. (1993). Increasing steps in recall of events: Factors facilitating immediate and long-term memory in 13.5- and 16.5-month-old children. *Child Development, 64,* 1204–1223.

Bauer, P. J., Hertsgaard, L. A., & Dow, A. (1994). After 8 months have passed: Long-term recall of events by 1- to 2-year-old children. *Memory, 2,* 353–382.

Bauer, P. J., & Mandler, J. M. (1990). Remembering what happened next: Very young children's recall of event sequences. In R. Fivush & J. A. Hudson (Eds.), *Knowing and remembering in young children,* (pp. 9–29). New York: Cambridge University Press.

Bauer, P. J., Wenner, J. A., Dropik, P. L., & Wewerka, S.S. (2000) Parameters of remembering and forgetting in transition from infancy to early childhood. *Monographs of the Society for Research in Child Development, 65,* v–204.

Bauer, P. J., & Wewerka, S. S. (1995). One- to two-year olds' recall of events: The more expressed, the more impressed. *Journal of Experimental Child Psychology, 59,* 475–496.

Bauer, P. J., & Wewerka, S. S. (1997). Saying is revealing: Verbal expression of event memory in the transition from infancy to early childhood. In P. van den Broek, P. J. Bauer, & T. Bourg (Eds.), *Developmental spans in event comprehension and representation: Bridging fictional and actual events,* (pp. 139–168). Hillsdale, NJ: Lawrence Erlbaum Associates, Inc.

Barr, R., Dowden, A., & Hayne, H. (1996). Developmental changes in deferred imitation by 6- to 24-month-old infants. *Infant Behavior and Development, 19,* 159–170.

Barr, R., & Hayne, H (1999). Developmental changes in imitation from television during infancy. *Child Development, 70,* 1067–1081.

Boller, K., Rovee-Collier, C., Borovsky, D., O'Conner, J., & Shyi, G. (1990). Developmental changes in the time-dependent nature of memory retrieval. *Developmental Psychology, 26,* 770–779.

Brainerd, C. J., Reyna, V. F., Howe, M. L., & Kingma, J. (1990). The development of forgetting and reminiscence. *Monographs of the Society for Research in Child Development, 55,* v–93.

Butler, J., & Rovee-Collier, C. (1989). Contextual gating of memory retrieval. *Developmental Psychobiology, 22,* 533–552.

Campbell, B. A., & Jaynes, J. (1966). Reinstatement. *Psychological Review, 73,* 478–480.

Deocampo, J. A., & Hudson, J. A. (2003). Reinstatement of 2-year-olds' event memory using photographs. *Memory, 11,* 13–25.

Fivush, R., & Hamond, N. R. (1989). Time and again: Effects of repetition and retention interval on 2-year-olds' event recall. *Journal of Experimental Child Psychology, 47,* 259–273.

Galluccio, L., & Rovee-Collier, C. (1999). Reinstatement effects on retention at 3 months of age. *Learning and Motivation, 30,* 296–316.

Greco, D., Rovee-Collier, C., Hayne, H., Griesler, P., & Earley, L. (1986). Ontogeny of early event memory: I. Forgetting and retrieval by 2- and 3-month-olds. *Infant Behavior and Development, 9,* 441–460.

Hartshorn, K. (2003). Reinstatement maintains a memory in human infants for 1½ years. *Developmental Psychobiology, 42,* 269–282

Hartshorn, K., Rovee-Collier, C., Gerhardstein, P., Bhatt, R. S., Wondoloski, T. L., Klein, P., et al. (1998). The ontogeny of long-term memory over the first year and a half of life. *Developmental Psychobiology, 32,* 69–89.

Hildreth, K., Sweeney, B., & Rovee-Collier, C. (2003). Differential memory-preserving effects of reminders at 6 months. *Journal of Experimental Child Psychology, 84*, 41–62.

Howe, M. L., Courage, M. L., & Bryant-Brown, L. (1992). Reinstating preschoolers' memories. *Developmental Psychology, 29,* 854–869.

Howe, M. L., Kelland, A., Bryant-Brown, L., & Clark, S. L. (1992). Measuring the development of children's amnesia and hyperamnesia. In M. L. Howe, C. Brainerd, & V. F. Reyna (Eds.), *Development of long-term retention* (pp. 56–102). New York: Springer-Verlag.

Hudson, J. A. (1990). The emergence of autobiographic memory in mother–child conversation. In R. Fivush & J. A. Hudson (Eds.), *Knowing and remembering in young children* (pp. 166–196). New York: Cambridge University Press.

Hudson, J. A. (1993). Reminiscing with mothers and others: Autobiographic memory in young two-year-olds. *Journal of Narrative and Life History, 3,* 1–32.

Hudson, J. A., & Sheffield, E. G.(1998). D!jà vu all over again: Effects of reenactment on toddlers' event memory. *Child Development, 69,* 52–67.

Klein, P. J., & Meltzoff, A. N (1999) Long-term memory, forgetting and deferred imitation in 12-month-old infants. *Developmental Science, 2,* 102–113.

Lemish, D., & Rice, M. L. (1986).Television as a talking picture book: A prop for language acquisition. *Journal of Child Language, 13,* 251–274.

Meltzoff, A. N. (1995). What infant memory tells us about infantile amnesia: Long-term recall and deferred imitation. *Journal of Experimental Child Psychology, 59,* 497–515.

Meltzoff, A. N. (1988). Imitation of televised models by infants. *Child Development, 59,* 1221–1229.

Meltzoff, A. N., & Moore, M. K. (1994). Imitation, memory, and the representation of persons. *Infant Behavior and Development, 17,* 83–99.

Nelson, K. (1993). The psychological and social origins of autobiographic memory. *Psychological Science, 4,* 1–8.

Rovee-Collier, C. (1995). Time windows in cognitive development. *Developmental Psychology, 31,* 147–169.

Rovee-Collier, C. (1999). The development of infant memory. *Current Directions in Psychological Science, 8,* 80–85.

Rovee-Collier, C., Griesler, P. C., & Earley, L. A. (1985). Contextual determinants of retrieval in three-month-old infants. *Learning and Motivation, 16,* 139–157.

Rovee-Collier, C. K., & Hayne, H. (1987). Reactivation of infant memory: Implications for cognitive development. In H. W. Reese (Ed.), *Advances in child development and behavior* (Vol. 20, pp. 185–238). San Diego: Academic.

Sheffield, E. G. (2004). But I thought it was Mickey Mouse: The effects of post-event information on 18-month-olds memory. *Journal of Experimental Child Psychology, 87,* 221–238.

Sheffield, E. G., & Hudson, J. A. (1994). Reactivation of toddlers' event memories. *Memory, 2,* 447–465.

Spear, N. E., & Parson, P. J. (1976). Analysis of a reativation treatment: Ontogenetic determinants of all eviated forgetting. In D. L. Medin, W. A. Roberts, & R. T. Davis (Eds.), *Process of Animal Memory* (pp. 135–165). Hillsdale, NJ: Lawrence Erlbaum Associates, Inc.

Copyright of Journal of Cognition & Development is the property of Lawrence Erlbaum Associates and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.

View publication stats